Incomplete form of Primary Hypertrophic Osteoarthropathy (Touraine-Solente-Gole Syndrome) Masquerading as Polyartrhalgia Diagnosed in Technetium-99m-Methylene Diphosphonate Scintigraphy: An Interesting Case Report

Abstract

The primary hypertrophic osteoarthropathy (PHOA) (pachydermoperiostosis) is a rare genetic/hereditary disease characterized by skin changes (pachydermia), clubbing of fingers and periosteal thickening (periostitis) with sub-periosteal new bone formation. Here we describe a case of an adolescent male who presented with clubbing and polyarthralgia. On evaluation with scintigraphy and SPECT-CT, he was diagnosed to have incomplete form of PHOA(skeletal manifestations without skin changes). The identification of incomplete form of primary hypertrophic osteoarthropathy which can be easily misdiagnosed as rheumatoid arthritis is discussed here.

Keywords: *Bone scan, clubbing, primary hypertrophic osteoarthropathy*

Thangalakshmi Sivathapandi, Jaykanth Amalachandran, Shelley Simon, Indirani Elangovan

Department of Nuclear Medicine and PET-CT, Apollo Hospitals, Chennai, Tamil Nadu, India

Introduction

Primary hypertrophic osteoarthropathy (PHOA) or pachydermoperiostosis or Touraine-Solente-Gole syndrome is a rare genetic disorder usually characterized by pachydermia (thickening of the skin), periostosis (excessive bone formation), and clubbing (swelling of tissue with loss of normal angle between nail and nail bed). It usually manifests in adolescence, occurring almost exclusively in males, with M:F ratio of 7:1. It is associated with significant morbidity with advancing age.[1,2] Three recognized forms of primary osteoarthropathy are (1) complete (pachydermia, digital clubbing, and periostosis), (2) incomplete (no pachydermia), and (3) fruste form (prominent pachydermia with few skeletal manifestations).[3,4] It has to be differentiated from other causes of hypertrophic periostitis as well as thyroid acropachy by clinical as well as imaging correlation.

Case Report

A 26-year-old male was referred to the nuclear medicine department by the rheumatologist with a 3-year history of polyarthralgia. The pain was initially involving knees, followed by the wrist and subsequently the ankles and small joints of hands and feet. The pain was progressive in nature and associated with swelling in joints. No history of early morning stiffness. He also noted some change in the shape of his nails since his childhood. Other symptoms to suggest chest or abdominal disease were absent. His father also had a history of nail changes, otherwise family history was unremarkable. No history of any dermatological problem.

On examination, Grade III clubbing was present in the fingers and toes [Figures 1 and 2]. The distal end of both forearms was swollen. No swelling was present elsewhere in the other joints. No clinical evidence of skin lesions noted. Examination of other systems was unremarkable.

On further investigations, his erythrocyte sedimentation rate was 34 mm in 1st h, and hemoglobin, leukocyte, and other cell counts were found to be normal. Liver function test was within normal limits. Human leukocyte antigen B-27, rheumatoid factor, and anti-cyclic citrullinated peptides were negative. X-ray chest and knee joint

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: reprints@medknow.com

How to cite this article: Sivathapandi T, Amalachandran J, Simon S, Elangovan I. Incomplete form of Primary Hypertrophic Osteoarthropathy (Touraine-Solente-Gole Syndrome) Masquerading as Polyartrhalgia Diagnosed in Technetium-99m-Methylene Diphosphonate Scintigraphy: An Interesting Case Report. Indian J Nucl Med 2018;33:128-31.

Address for correspondence:
Dr. Thangalakshmi Sivathapandi, Department of Nuclear Medicine and PET-CT, Apollo Hospitals, No. 21, Greams Lane, Off Greams Road, Chennai - 600 006, Tamil Nadu, India.
E-mail: tlakshmiraj2012@gmail.com

Access this article online

Website: www.ijnm.in

DOI: 10.4103/ijnm.IJNM_134_17

Quick Response Code:

© 2018 Indian Journal of Nuclear Medicine | Published by Wolters Kluwer - Medknow



Figure 1: Image showing Grade 4 clubbing (Drumstick appearance) of digits



Figure 2: Image showing Grade 4 clubbing (Drumstick appearance) of the digits

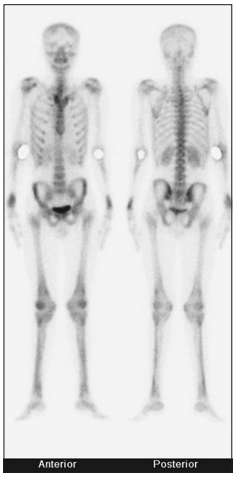


Figure 3: Tc99m methylene diphosphonate whole body bone scan shows diffuse increase in tracer concentration along the cortical margins of bilateral radius and ulna (predominantly epiphyseal region), tibiae, fibulae, and femoral



Figure 4: Tc99m methylene diphosphonate static images of hands and feet shows increased tracer uptake in the distal radius, distal ulna, distal tibia, distal phalanges of hands, metacarpal, metatarsal, and intertarsal regions

findings were normal. Echocardiogram study showed no evidence of pulmonary artery hypertension.

Skeletal scintigraphy was performed 3 h after the administration of 740 MBq/20 mCi of Tc-99m methylene diphosphonate (MDP). Planar images showed diffusely increased tracer concentration in the distal third of bilateral radius and ulna with expansion of epiphyseal region, symmetrical increased tracer concentration along the cortical margins of bilateral distal femorae, entire shaft of tibiae and fibulae [Figure 3]. Increased tracer concentration is also seen in the distal phalanges, wrists, MCP joints of both hands, metatarsal and intertarsal regions of both feet [Figure 4]. Single-photon emission computed tomography (CT) with low-dose CT of the wrist was done which showed increased tracer concentration along the entire width of distal third of bilateral radius and ulna. Corresponding CT shows evidence of periosteal thickening bilaterally with areas of minor lucency in the left distal radius suggesting possible acrolysis [Figure 5]. Increased tracer concentration is also seen in distal phalanges with soft-tissue swelling (consistent with clubbing).

Discussion

PHOA is a rare syndrome with clubbing of the digits, periostitis of the bones, and arthritis.[5] Pachydermoperiostosis a complete form of primary osteoarthropathy is a rare familial autosomal dominant disorder with no known etiology and accounts for 5% of all the cases of hypertrophic osteoarthropathy.[6,7] PHOA has been described in three forms, namely, classic or complete form, with skin and skeletal changes; incomplete form, with skeletal changes but no skin changes; and forme fruste with skin changes but no skeletal findings.[8]

Our patient had only skeletal manifestations with no skin changes, so he falls under incomplete form of PHOA. Skeletal findings of PHOA usually includes symmetric periosteal thickening (new bone formation) in the diametaphyseal and epiphyseal region of long bones,

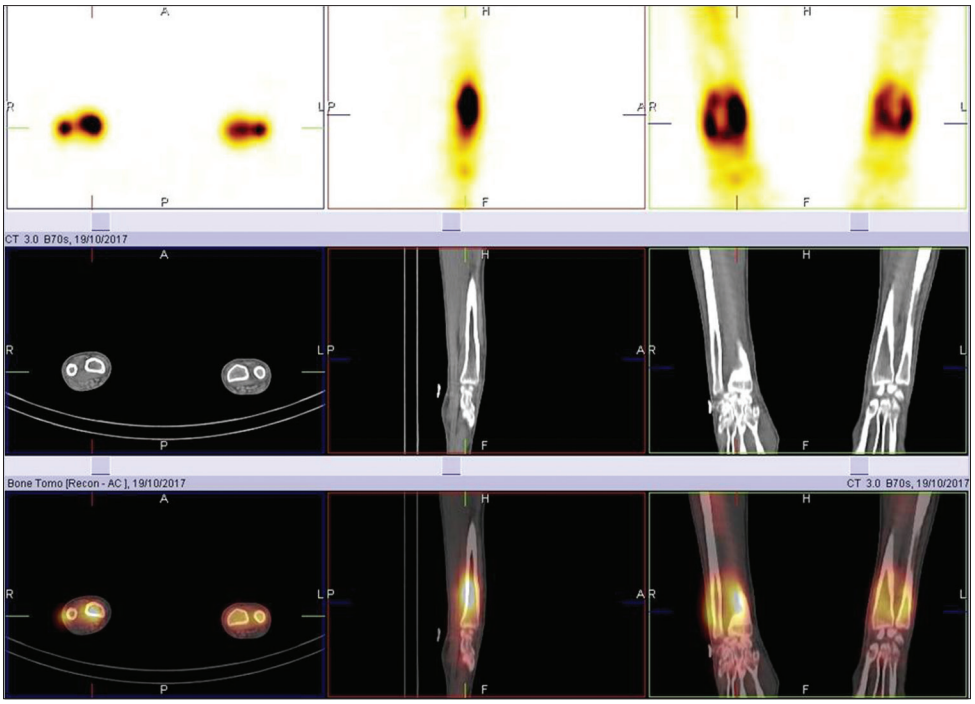


Figure 5: Single-photon emission computed tomography-computed tomography images of forearm shows diffusely increased tracer concentration along distal part of radius and ulna with evidence of some lysis in distal radius

especially of the forearm and leg, but can also involve the metacarpals, metatarsals, and phalanges. Involvement of the epiphyseal region distinguishes it from the secondary form, in which the epiphyses are usually spared.[9]

Early diagnosis helps in reducing the morbidity and in prognostication. Tc-99m MDP bone scan is a very sensitive investigation for detecting PHOA at early stage. A diagnosis of primary osteoarthropathy must be considered in a patient of adolescent age group presenting with clubbing, joints pain, and symmetrical tracer uptake along the cortical margins of long bones of legs and forearms with predominant epiphyseal involvement on bone scintigraphy.[10]

Uptake in the small tubular bones of the hands and feet may be attributed to the vasodilatation in the tips of fingers, excessive growth of cellular tissue in the clubbed fingers and the duration of disease. A complex interplay of these factors has been suggested for increased activity of the radiotracer in the phalanges.[11] Due to the benign course of PHOA, it is important to distinguish it on scintigraphy from diseases such as secondary hypertrophic osteoarthropathy and thyroid acropachy. In thyroid acropachy, the uptake was more in the diaphyseal region usually confined to the small bones of hands and feet and rarely involving long bones.[12,13]

Conclusion

In patients presenting with joint pain and clubbing without dermal findings and showing symmetrical increased uptake along the distal end of long bones and in small bones of hands and feet on bone scintigraphy, the diagnosis of incomplete form of PHOA should be considered.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form the patient(s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

References

1. Goyal S, Schwartz RA, Richards GM, Goyal R. Pachydermoperiostosis. Available from: http://www.emedicine.com. [Last accessed on 2006; Last updated on 2008 Apr 30].
2. Sahasrabhojaney VS, Hinge AV, Ghodeswar SS, Machnurkar AS, Daware AM, Nagarik AP. Touraine-Solente-Gole' syndrome. J Indian Acad Community Med 2005;6:152-4.
3. Touraine A, Solente G, Gole L. Pachydermoperiostosis (Touraine-Solente-Golé syndrome). Case report. Presse Med 1935;43:1820-4.
4. Karnan S, Krishnamoorthy V, Ethiraj P, Sathyanathan BP. Touraine-Solente-Gole syndrome: The complete form needs to be recognized. Indian J Nucl Med 2012;27:201-4.
5. Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL,

Loscalzo J, editors. Harrison's Principles of Internal Medicine. New York: McGraw-Hill Medical; 2008.

6. Latos-Bielenska A, Marik I, Kuklik M, Materna-Kiryluk A, Povysil C, Kozlowski K, *et al.* Pachydermoperiostosis-critical analysis with report of five unusual cases. Eur J Pediatr 2007;166:1237-43.
7. Eckardt A, Kreitner KF. Primary hypertrophic osteoarthropathy. Z Orthop Ihre Grenzgeb 1995;133:303-5.
8. Bhaskaranand K, Shetty RR, Bhat AK. Pachydermoperiostosis: Three case reports. J Orthop Surg (Hong Kong) 2001;9:61-6.
9. Resnick D. Enostosis, hyperostosis, and periostitis. In: Resnick D, Kransdorf MJ, editors. Bone and Joint Imaging. 3rd ed. Philadelphia: Elsevier Saunders; 2005. p. 1433-5.
10. Mudalsha R, Jacob M, Jora C, Pandit A. Tc-99m MDP bone scintigraphy in a case of Touraine-Solente-Gole syndrome. Indian J Nucl Med 2011;26:46-8.
11. Bomanji J, Nagaraj N, Jewkes R, Fields M, Maini RN. Pachydermoperiostosis: Technetium-99m-methylene diphosphonate scintigraphic pattern. J Nucl Med 1991;32:1907-9.
12. Seigel RS, Thrall JH, Sisson JC. 99mTc-pyrophosphate scan and radiographic correlation in thyroid acropachy: Case report. J Nucl Med 1976;17:791-3.
13. Gimlette TM. Thyroid acropachy. Lancet 1960;1:22-4.